

NCERT Multilingual Knowledge Hub

Requirements Specification Document

1. Purpose

The purpose of this system is to bridge the gap between English-dominated educational resources and the needs of regional language students of grade 6-12 by providing a highly structured, multilingual knowledge synthesis and content delivery system. Grounded in the NCERT curriculum, the system serves as a digital encyclopedia that provides high-quality STEM (Science and Mathematics) education in Telugu, Hindi, and Odia.

The system is designed to:

- **Universalize Access:** Provide native-language students with the same depth of "Enriched" content usually found in English-only resources.
- **Ensure Accuracy:** Utilize a "**Generation-then-Translation**" pipeline to maintain the technical integrity of scientific concepts across languages. We have adopted this approach because current LLMs often struggle with high-quality, technically accurate content generation directly in regional languages; therefore, we will generate the primary content in English and leverage specialized translation models to ensure the Hindi, Telugu and Odia versions are correct.
- **Enable Scalable Education:** Create a structured knowledge repository that eliminates the latency and hallucinations associated with real-time LLM interactions.

2. Scope of the System

The project focuses on creating a "Source-of-Truth" educational platform. Unlike general AI chatbots that can hallucinate, this system is grounded in the NCERT curriculum for Science and Mathematics.

In Scope:

- **Hierarchical Navigation:** A rigid organizational structure following **Grade > Subject > Chapter > Topic**.
- **Multilingualism:** Native-quality Hindi, Telugu and Odia content generated via a specialized translation pipeline.
- **Mathematical Precision:** Rendering of complex formulas and equations using LaTeX/KaTeX across all supported languages.

Out of Scope:

- **Real-time Content Generation:** To ensure good quality and no latency, the system does not generate content "on-the-fly" during user interaction.
- **Student Assessment/Grading:** The system is a knowledge repository, not a platform for submitting or grading student assignments.

3. Functional Requirements

3.1 Hierarchical Data Management

- **3.1.1 Structural Integrity:** The system shall organize all knowledge assets into a strict four-tier hierarchy: **Grade > Subject > Chapter > Topic**.
- **3.1.2 Granular Topic Mapping:** Each chapter must be decomposed into individual topics (sub-sections) to ensure that content is digestible and easy for students to navigate.

3.2 Automated Content Generation Pipeline

- **3.2.1 Source Grounding:** The system shall use NCERT PDF text as the primary "Source of Truth" for all content generation along with some secondary trusted sources to prevent hallucinations.
- **3.2.2 Contextual Enrichment:** The pipeline shall supplement textbook facts with simplified analogies, examples or real-world applications etc. fetched from verified external sources.
- **3.2.3 Synthesis of Information:** The system shall synthesize multiple sources into a consistent format, maintaining a pedagogical tone appropriate for the specific grade level.

3.3 Multilingual Processing

- **3.3.1 English-Master Workflow:** The system shall first generate a finalized "Master" version in English before initiating any translation processes.
- **3.3.2 Technical Translation:** The system shall translate the English Master into Hindi, Telugu and Odia using specialized models to ensure the accuracy of scientific and mathematical terminology.
- **3.3.3 Format Preservation:** The translation pipeline must preserve non-translatable elements, including LaTeX formulas and technical identifiers, across all language versions.

3.4 Content Delivery and UI/UX

- **3.4.1 Language Switching:** The user interface shall provide a seamless toggle to switch between Telugu, Hindi, and Odia.
- **3.4.2 Static Content Serving:** The system shall fetch and display pre-generated content from the database, ensuring zero-latency response times for the end-user.

4. Proposed Models (Under Evaluation)

We intend to explore several state-of-the-art models to find the best fit for our platform. We will try multiple models and adopt the one that provides the highest accuracy for technical terminology while keeping the language simple and easy for students to understand.

- **For Content Synthesis (English):**
 - **Llama 3.1/3.2 (3B/8B):** We will try these for their strong reasoning capabilities and ability to simplify complex Physics and Math concepts.
 - **Qwen 3 (4B/8B):** We might test these as an alternative to evaluate their performance in structured content generation and logical consistency.
- **For Translation (The Multilingual Core):**
 - **IndicTrans2 (AI4Bharat):** This is a primary candidate due to its specialized training on Indian languages and technical datasets.
 - **T5Gemma / TranslateGemma:** We will try this family of open-weights models from Google. Since it is optimized for translation efficiency, we will evaluate if it handles Telugu, Odia, and Hindi scientific contexts more fluently than traditional models.

We plan to test multiple combinations of these models to see which one delivers the best overall result. Our goal is to ensure that the final output provide precise scientific data, and easy explanations. The output will focus on contextual ease and daily-use vocabulary instead of rigid formal translation. This ensures the explanations in Hindi, Telugu, and Odia are relatable, intuitive, and easy for students to follow naturally.

5. Proposed Architectures (Experimental)

We will evaluate the following architectural frameworks to ensure the system is accurate, scalable, and easy for students to use.

5.1 Topic-Driven RAG and Translation Pipeline (Primary Focus)

This is our main pipeline, which uses a Vector Database to ground the AI in textbook facts before translating them.

- **Step 1 (Extraction):** We will parse the NCERT Index to create a master topic list.
- **Step 2 (Vector Retrieval):** The textbooks will be chunked and stored in a **Vector DB** . For every topic, we will retrieve the most relevant chunks to act as the "Source of Truth."
- **Step 3 (Web Enrichment):** We will try using search APIs to fetch extra content to supplement the textbook data.
- **Step 4 (Synthesis):** An LLM will combine the textbook chunks and web data to generate a simplified content in English.
- **Step 5 (Translation):** We will pass the English text through IndicTrans2 or T5Gemma for translation.
- **Step 6 (Storage):** The final content will be saved in PostgreSQL for low latency delivery to the student.

5.2 Offline Batch Knowledge Compilation Architecture (Non-RAG)

In this framework, the system does not use RAG at serving time or during synthesis.

- **Step 1 (Extraction):** We will parse the NCERT Index to create a master topic list.
- **Step 2 (Context Feeding):** Instead of isolated chunks, the system is fed the full relevant NCERT topic content along with selected trusted references.
- **Step 3 (Synthesis):** The model generates a single, carefully curated English master article per topic.
- **Step 4 (Translation):** We will pass the English text through IndicTrans2 or T5Gemma for translation.
- **Step 5 (Storage):** The final content will be saved in PostgreSQL for low latency delivery to the student.

The following designs are under consideration for the current phase; we will try to experiment with more architectures and will use the best of them.

6. Future Works

While the current phase focus is on building a robust multilingual knowledge hub, we have identified the following areas for future development and experimental integration:

6.1 Advanced Multi-Modal Content Integration

- **Automated Flowchart Generation:** We will continue to experiment with utilizing LLMs to generate accurate **Mermaid.js** code for process diagrams and flowcharts. Since technical accuracy in diagrams is critical, future work will focus on ensuring these non-textual representations meet strict pedagogical standards.
- **Contextual Video Mapping:** Since high-quality regional language videos for specific topics are often difficult to find, we will evaluate the feasibility of automatically identifying and mapping native-language video resources to their respective topics.